Science of the Total Environment 607–608 (2017) 262–270

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Side-chain fluorinated polymer surfactants in aquatic sediment and biosolid-augmented agricultural soil from the Great Lakes basin of North America

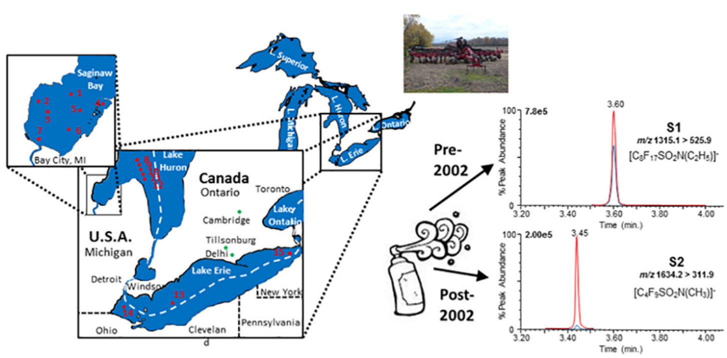
Shaogang Chu, Robert J. Letcher *

Wildlife and Landscape Science Directorate, Science and Technology Branch, Environment and Climate Change Canada, National Wildlife Research Centre, Carleton University, Ottawa, ON K1A 0H3, Canada

HIGHLIGHTS

- Major side-chain fluoroalkyl components found in pre- and post-2002 Scotchgard products
- High concentrations of the pre-2002 product component found in biosolid-augmented agricultural soil samples
- Both components were at greater concentrations in soil and sediment samples than all other perfluorinated compounds
- These results help to explain the low PFAS portion accountability in extractable organic fluorine content in sediments

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 31 March 2017
Received in revised form 27 June 2017
Accepted 28 June 2017
Available online xxxx

Editor: Kevin V. Thomas

Keywords:
Perfluoroalkyl substances
PFASs
Scotchgard™ fabric protector
WWTP
LC-MS/MS

ABSTRACT

Side-chain fluorinated polymer surfactants are the main components of fabric protector sprays and used extensively on furniture and textiles. The composition of these commercial protector products has changed, but there is currently a total dearth of information on these novel fluorinated polymers in the environment. Using a developed analytical approach, two complementary studies examined the distribution of Scotchgard™ fabric protector components in aquatic sediment and in agricultural soils where wastewater treatment plant (WWTP) sourced biosolid application occurred, and in samples from sites in the Laurentian Great Lakes basin of North America. The main components in the pre- and post-2002 Scotchgard™ fabric protectors were identified by MS/MS and Q-TOF-MS to contain a perfluorooctane sulfonamide (S1) and perfluorobutane sulfonamide (S2) based side-chain, respectively, and bonded to a polymer backbone. In fifteen sediment samples collected in 2012–2013 from western Lake Erie and Saginaw Bay (Lake Huron), S1 was in all sediment samples (0.18 to 461.59 ng/g dry weight (d.w.)); S2 was in 80% of the sediment samples (<0.03 to 24.08 ng/g d.w.). Thirteen soil samples were collected (2015) from a biosolid applied and two non-biosolid applied farm field sites in southern Ontario (Canada). S1 was detected in 100% of the soil samples from biosolid-augmented agricultural sites (mean 236.36 ng/g d.w.; range 41.87 to 622.46 ng/g d.w.), and at concentrations much greater than in the aquatic sediment samples. The concentration of S1 and S2 in soil and sediment samples were also much greater than the total concentration of other per-and poly-fluoroalkyl substances (PFASs) that were measured.

* Corresponding author.
E-mail address: robert.letcher@canada.ca (R.J. Letcher).

http://dx.doi.org/10.1016/j.scitotenv.2017.06.252

The ratio of S1 concentration versus $\sum_{22}$PFAS concentration was up to 1616 in sediment samples from Lake Erie. This results helps to explain why known PFASs account for low percentages of the total extractable organic fluorine (EOF) content in sediment.

1. Introduction

Per and poly-fluoroalkyl substances (PFASs) including their precursors have been produced since mid-20th century and used in numerous industrial and commercial applications (Prevedouros et al., 2006; Lindstrom et al., 2011; Buck et al., 2011; Betts, 2007). Their widespread use has resulted in a release of PFASs into the environment either from direct sources as well as from indirect sources, such as via degradation of precursors (Prevedouros et al., 2006; Lindstrom et al., 2011; Zareitalabad et al., 2013; Benskin et al., 2012; Nadal and Domingo, 2014; Anderson et al., 2016; Liu and Avendano, 2013; Houtz et al., 2013). Worldwide investigations have shown that PFASs, particularly perfluorinated carboxylic acids (PFCAs) and sulfonic acids (PFSAs), are environmentally ubiquitous and including in humans (Buck et al., 2011; Ahrens et al., 2009; Rankin et al., 2016; Kwadijk et al., 2010; Labadie and Chevreuil, 2011; Letcher et al., 2015; Butt et al., 2010; Olsen et al., 2005; Olsen et al., 2012; Braune and Letcher, 2013; Houde et al., 2006; Houde et al., 2011; White et al., 2015).

PFAS studies have mostly targeted PFCAs (C_nF_{2n} + COOH), PFSAs (C_nF_{2n} + SO_3H) and some of their possible precursors, which are of low molecular weight. However, there has been very few studies on larger PFAS functional derivatives and polymers (Chu and Letcher, 2014; Russell et al., 2008; Washington et al., 2009). Side-chain fluorinated polymers are important industrial materials, which possess polyfluorinated chains that are chemically bonded to non-fluorinated polymer backbones (Buck et al., 2011). Side-chain fluorinated polymers represent a high percentage of all commercially available PFAS products and are used in a variety of household products such as fabric protector sprays (Buck et al., 2011; Liu and Avendano, 2013; Wang et al., 2014).

The largest producer of PFOS and its C_8 chemistry was The 3 M Company, which phased-out its PFOS-based production in 2002 (The 3M Company, 2000). For the first time, to our knowledge, we recently reported that pre- and post-2002 Scotchgard™ fabric protectors contain side-chain fluorinated polymers that could be metabolized to perfluoroalkyl sulfonamides in vitro using a rat liver microsomal based assay (Chu and Letcher, 2014). These fluorinated polymer components are widely used in commercial applications and possess strong sorption behaviour on surfaces in human indoor environments (e.g. textiles, upholstery) (Renner, 2006). It can be hypothesized that these side-chain fluorinated polymers are present in the environment, especially in receiving compartments from wastewater treatment plant (WWTP) outflows, such as sediment, sludge and soil, and result in important indirect pollution sources of PFASs (Arvaniti and Stasinakis, 2015; Buck et al., 2011; Lindstrom et al., 2011).

There are uncertainties surrounding the routes and magnitude of PFAS contamination in terrestrial soils and aquatic sediments (Prevedouros et al., 2006; Wang et al., 2014). Codling et al. (2014) measured total fluorine (TF), extractable organic fluorine (EOF) and poly- and per-fluorinated compounds (abbreviated PFCs rather than the correct PFASs) in some dated sediment cores and grab surface sediments from Lake Michigan. Of the total elemental fluorine, only 0.2% was accounted for by quantifiable PFCs in the sediment cores and surface samples. A previous study in Lake Ontario also hypothesized that significant amounts of unidentified organic fluorine might present in sediment, and this unidentified organic fluorine could include precursor and/or intermediate compounds of PFCAs and PFSAs (Yeung et al., 2013).

At present little is known about behaviour of side-chain fluorinated polymers in the environment (Buck et al., 2011; Liu and Avendano, 2013; Russell et al., 2008). In this study, using a novel developed analytical method, which included a d-SPE cleanup step and UHPLC/MS/MS determination, we investigated and compared the main components of Scotchgard™ fabric protectors, i.e. side-chain fluorinated polymer surfactants, in aquatic sediment samples and in agricultural soil samples that had been augmented with WWTP-sourced biosolids from study sites in the southern half of the Laurentian Great Lakes basin of North America.

2. Materials and methods

2.1. Chemicals and materials

Scotchgard™ pre-2002 formulation (Tech. mix) (100 μg/mL in methanol) and Scotchgard™ post-2002 formulation (Tech. mix) (100 μg/mL in methanol) were purchased from AccuStandard Inc. (New Haven, CT, USA). As detailed by the supplier, these two standard solutions were prepared from commercial products of Scotchgard™ fabric protector produced before 2002 and after 2002 by The 3 M Company. Working standard solutions (with concentration of 100 ng/mL and 1000 ng/mL) were prepared by mixing these two standard solutions and then diluting with methanol. Perfluoro-1-butane-sulfonamide (FBSA) was synthesized and purified in our lab according the method described by Benfodda et al. (2010) with some modifications as detailed in Chu et al. (Chu and Letcher, 2014). Perfluoro-1-ethylcyclohexylsulfonate (PFEtCHxS) was purchased from Campro Scientific GmbH (Berlin, Germany). All other PFAS standard solutions, including the mass-labelled internal standards, were purchased from Wellington Laboratories (Guelph, ON, Canada). The standard solutions were prepared according our previously published paper (Chu et al., 2016).

Disposable silica gel solid-phase extraction (SPE) columns (500 mg; J.T. Baker®) were purchased from VWR International (Mississauga, ON, Canada). Oasis WAX SPE cartridges (60 mg × 3 mL) were purchased from Waters Limited (Mississauga, ON, Canada). Supel QuE *Z*-Sep sorbent was purchased from Sigma-Aldrich (Oakville, ON, Canada). PVDF MicroSpin centrifuge filters (0.20 μm, 500 μL) was purchased from Fisher Scientific (Ottawa, ON, Canada).

Ammonium hydroxide solution (28–30%), Ultra CHROMASOLV grade methanol (MeOH), acetonitrile (ACN), hexane and dichloromethane (DCM) were purchased from Sigma-Aldrich. Ultrapure water was obtained from a Milli-Q system. All other reagents were the highest commercial purity and used as received. All glassware used in sample extraction and cleanup process was treated at 450 °C overnight in box furnace to destroy any possible organic contamination.

2.2. Great Lakes sample collection and preparation

The collection and sampling of all Laurentian Great Lakes sediment samples was detailed fully in our previously published paper (Trouborst et al., 2014), and listed in Table 1 and Fig. 1. Briefly, surficial sediment samples were collected by colleagues from Environment and Climate Change Canada and the United States Environmental Protection Agency (Great Lakes Environmental Center (GLEC)). Briefly, sampling was conducted using a 15 × 15 cm Petite Ponar Grab (stainless steel box) sampler (EcoEnvironmental, Perth, WA, USA), where a top 6 cm sample of surficial bottom sediment was collected. Grab samples were collected from across Lake Erie (n = 3; western, central and eastern basins) in May 2012 (n = 1) and in August 2013 (n = 2). The GLEC

Table 1
Sediment sampling site locations in Lake Erie and Lake Huron of the Laurentian Great Lakes in Canada and the U.S.A., and agricultural soil sampling sites located in southern Ontario, Canada.

Sample	Location	Latitude	Longitude	Depth (m)	Sampling date	Substrate
Sediment[a]						
1	Saginaw Bay, Lake Huron	43.9417	−83.62390	11.6	12-Sep-13	Silt
2	Saginaw Bay, Lake Huron	43.8953	−83.86060	3	12-Sep-13	Fine sand
3	Saginaw Bay, Lake Huron	43.8381	−83.79280	7	12-Sep-13	Silt
4	Saginaw Bay, Lake Huron	43.8833	−83.39310	2.7	12-Sep-13	Silt
5	Saginaw Bay, Lake Huron	43.8469	−83.56250	2.7	12-Sep-13	Medium sand
6	Saginaw Bay, Lake Huron	43.7383	−83.64080	3	12-Sep-13	Fine sand
7	Saginaw Bay, Lake Huron	43.6847	−83.84310	3.3	12-Sep-13	Fine sand
8	Lake Huron	44.3325	−82.8503	67	07-Aug-13	Silt
9	Lake Huron	44.2711	−82.8133	64	08-Aug-13	Silt
10	Lake Huron	44.2087	−82.7914	59	09-Aug-13	Silt
11	Lake Huron	44.1464	−82.7703	50	10-Aug-13	Silt
12	Lake Huron	44.0815	−82.746	37	11-Aug-13	Medium sand
13	Lake Erie	41.8727	−82.0432	8.7	16-May-12	Silt
14	Lake Erie	41.8847	−83.1656	N/A	01-Aug-13	Silt
15	Lake Erie	42.7614	−79.2053	N/A	02-Aug-13	Silt
Soil						
TI (n = 9)	Tillsonburg, ON, Canada	42.54	−80.43	Surface	May20–June13, 2014	Soil
DE (n = 1)	Delhi, ON, Canada	42.53	−80.14	Surface	02-Jun-14	Soil
CB (n = 3)	Cambridge, ON, Canada	43.22	−80.23	Surface	08-Jul-14	Soil

[a] See the map in Fig. 1 for the exact locations of the sampling sites.

collected sediment samples in August 2013 from Lake Huron open water sites (n = 5) and in September 2013 from Saginaw Bay sites (n = 7).

Thirteen soil samples were collected in the spring of 2014 from three agricultural field locations in southern Ontario, Canada in the Great Lakes basin (Table 1, Fig. 1). For the Cambridge (CB) site samples (n = 3), dewatered biosolid product from a secondary wastewater treatment plant had been applied to the field in the fall of 2013. Soil samples were also taken from fields near Tillsonburg and Delhi, Ontario which had received no known biosolid application.

All collected sediment and soil samples were stored in amber jars and frozen for transport to our lab (Organic Contaminants Research Laboratory, National Wildlife Research Centre, ECCC, Carleton University, Ottawa, ON, Canada). All soil and sediment samples were air-dried in a laboratory fume hood at room temperature. After that the samples were ground and sieved by 60 mesh sieve to remove stone and other matter, and the samples were stored in amber jars at 4 °C until sample analysis. A soil sample from agriculture field in Canada Agriculture and Food Museum in Ottawa (Ontario, Canada) was used as matrix for PFAS analysis method validation.

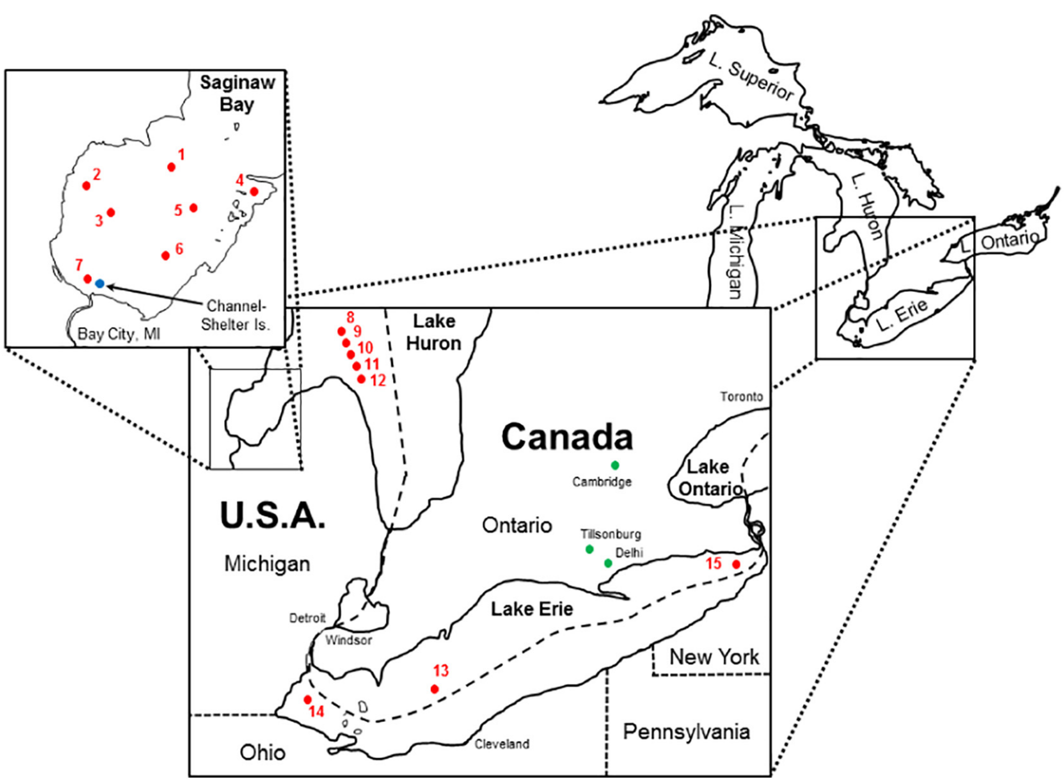


Fig. 1. Sampling sites (red dots) of surficial sediments from Lakes Erie and Huron in the Laurentian Great Lakes of North America. The insert shows details of the sampling sites in Saginaw Bay. Sampling sites of the soil samples from agricultural fields in southern, Ontario are also shown (green dots). See Table 1 for sample and sampling details. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The moisture in sediment and soil sample was measured gravimetrically by drying the sample in oven at 120 °C overnight. The concentration of target PFAS compounds in the sample was reported on a dry weight (d.w.) basis.

2.3. Sample extraction and clean up for analysis of side-chain fluorinated polymers in soil and sediment

For analysis of the Scotchgard™ fabric protector components in sediment and soil samples, 2 g of sample were weighted and placed in a glass culture tube. An 8 mL volume of the extraction solution acetone/hexane (50v/50v) was added and the sample was mixed by vortex for 1 min. The sample was then extracted by ultrasonication for 15 min in a sonication bath at 45 °C. After centrifugation, the supernatant solution was decanted into another glass culture tube. This process was repeated twice more and all the above extracts were combined. After the solvent in the extract was evaporated with a gentle flow of nitrogen, the residual was reconstituted in 2 mL DCM by vortex and then sonicated for 20 min.

The sample was first cleaned up using a disposable silica gel SPE column (500 mg). The column was conditioned with 3 mL DCM. After the sample solution was loaded on the column, the column was washed with 6 mL of DCM and then the target compounds were eluted out from the column with 4 mL of ACN/DCM (40v/60v) solvent. This fraction was collected and concentrated to dryness with a gentle flow of nitrogen and reconstituted in 2 mL of ACN. To reduce chemical interferences, the sample was then cleaned up using a dispersive solid-phase extraction (d-SPE) approach. A 200 mg amount of Supel QuE Z-Sep sorbent was added into the sample solution and then mixed by vortex for 1 min. The sample including sorbent was transferred into a PVDF MicroSpin centrifuge filter. After centrifugation the filtrate was transferred into a centrifuge tube and concentrated to dryness. The sample was then reconstituted in 50 μL of MeOH. A standard addition method was used for quantitative analysis of side-chain fluorinated polymers, where a 20 μL aliquot of the sample solution was transferred into a sample vial and spiked with 20 μL of MeOH. Another 20 μL aliquot of the sample solution was transferred into a another sample vial and spiked 20 μL of the Scotchgard™ fabric protector mixture standard solution (pre-2002 and post-2002) with a concentration of 100 ng/mL or 1000 ng/mL in MeOH. The samples were then ready for ultra-performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) analysis.

2.4. Analysis of Scotchgard™ fabric protectors by UHPLC/MS/MS and HPLC/Q-TOF

Quantitative analysis of the components of the was performed according to a previous study (Chu et al., 2016) and using a Waters XEVO-TQ-S UHPLC-QQQ-MS/MS (Waters Limited, Milford, MA, USA). Chromatographic separation was accomplished by a Kinetex® C8 LC Column (1.7 μm particle size, 100 Å, 2.1 × 50 mm, Phenomenex, CA, USA). The column and sampler temperature were 50 °C and 20 °C, respectively. The mobile phases were water (A) and methanol (B), both containing 2 mM of ammonium acetate. The flow rate was 0.6 mL/min. The gradient started at 5% B and increased to 95% B in 3 min, and was then held for 3 min. Thereafter, the mobile phase composition was returned to initial conditions and the column was allowed to equilibrate for 6 min.

The triple quadruple tandem mass spectrometer XEVO-TQ-S with an electrospray ionization source (ESI) was operated in the negative mode. Nitrogen was used as nebulizing gas and desolvation gas, and argon was used as collision gas. The capillary voltage was 1.5 KV. The desolvation temperature and cone temperature were 500 °C and 150 °C, respectively. Desolvation and cone gas flow rates were 800 and 150 L/h, respectively. Analyses were performed using the multiple reaction monitoring (MRM) mode, and the transitions, compound dependent operation parameters and retention times are listed in Table 2. The instrument control was via Waters Masslynx V4.1 software.

Because the lack of molecular structure information for the target compounds, finding a suitable internal standard was not possible. Therefore, a standard addition method was used to measure the concentration of side-chain fluorinated polymers to eliminate severe ESI(-) matrix effects for the analytes in sample. In this method, for each sample this was according to the equation:

$$C_x/(C_x + C_s) = A_x/A_{(x+s)}$$

where C_x and $(C_x + C_s)$ are the concentrations of the analyte without and with the standard addition, respectively, A_x and A_{x+s} are the chromatogram peak areas of the solutions containing C_x and $(C_x + C_s)$, respectively. The reported concentrations were normalized concentrations depending on the authentic standards by the supplier due to the lack of pure chemical standard.

An Agilent 1200 LC system coupled to an Agilent 6520A liquid chromatography-quadrupole time of flight mass spectrometer (LC-Q-TOF-MS) system (Agilent Technologies, Mississauga, ON, Canada) was used for identification of components of Scotchgard™ fabric protectors in standard solution and confirmation the results of measured components in the environment soil and sediment samples. The LC-Q-TOF-MS operation parameters were fully described in our previously published paper (Chu and Letcher, 2014).

2.5. 2.5. Analysis of other PFASs in sediment and soil samples

The analysis method for other PFASs, which includes PFCAs (13 compounds), PFSAs (5 compounds) and perfluoroalkyl sulfonamides (4 compounds), was fully described in the SI of our published paper (Chu et al., 2016) with some modifications for extraction of these compound from soil and sediment sample. The extraction was performed as that descripted above for analysis of Scotchgard™ fabric protectors components with solvent of 0.2 % formic acid in ACN.

2.6. QA/QC and method performance

The method limit of detection (MLODs) and method limit of quantification (MLOQ) for the components of the Scotchgard™ fabric protectors were defined as the concentration of target compounds in spiked agricultural soil sample producing a peak in chromatogram with a signal-to-noise (S/N) ratio of equal to 3 and 10 (peak-to-peak), respectively. In order to check for any blank contamination, a method blank sample was included in every batch of samples. Recoveries of the target analytes were assessed by five replicate analysis of spiked agricultural

Table 2
Compound dependent operation parameters for analysis of the main components (S1 and S2) in the Scotchgard™ fabric protectors by UHPLC-MS/MS.

Product	Main component	Precursor ion (*m/z*)	Product ion (*m/z*)	Cone voltage (V)	Collision energy (eV)	RT (min)
Scotchgard pre-2002	S1	1315.1	525.9	23	31	3.67
	S1 (confirm)	1315.1	744	8	18	
Scotchgard post-2002	S2	1634.2	311.9	10	41	3.51
	S2 (confirm)	1634.2	219	13	26	

soil sample. A standard addition approach was used to test for matrix effects from the agricultural soil sample (Matuszewski et al., 2003).

3. Results and discussion

3.1. Method development, optimization and validation

During the method optimization process for analysis of the Scotchgard™ fabric protector components, different extraction methods, extraction solvents and instrument operation parameters were validated in order to provide the best recovery and the least background interference. We first tried to extract target compounds from the soil and sediment samples using a Dionex ASE 200 accelerated solvent extraction system and different solvents including DCM, hexane, acetone and ACN. The sample was ground with diatomaceous earth (Agilent, Mississauga, ON, Canada) and extraction was performed at 150 °C under pressure of 1500 psi with a total three static cycles and flush of solvent equal to 60% of the cell volume. For the target compounds, accelerated solvent extraction method resulted in serious carryover contamination as a result of the instrument solvent system and the strong adsorption of the target compounds to solids. Moreover, in the ASE system some parts of the instruments are made of Teflon and other fluorinated polymers, such as tubing and seals, which might result in contamination and interference of the PFASs analysis.

An alternative ultrasonication extraction method was tested, using different extraction solvents including MeOH, 1% formic acid in MeOH, ACN, 1% formic acid in ACN, acetone, DCM, hexane/acetone (50v/5v). The extraction recoveries of target components from spiked agricultural soil samples using different solvents are shown in Table 3. As shown by the recovery efficiencies, the main component in the post-2002 Scotchgard™ fabric protector (S2) strongly adsorbed and was therefore difficult to extract from the soil matrix regardless of the solvent used. In contrast, the main component in the pre-2002 Scotchgard™ fabric protector (S1) was relatively easy to extract from the soil and with good recovery. The best recovery results were obtained using acetone/hexane (50v/50v) as the extraction solvent (Table 3).

The analysis of the Scotchgard™ fabric protector components in environmental samples was a challenge with respect to the sample cleanup as their molecular structures could not be completely determined. Although gel permeation chromatographic (GPC) separation is a very effective sample cleanup approach, and widely used in environment sample preparation and cleanup, GPC was found not to be suitable for the sample cleanup and analysis of the Scotchgard™ fabric protector components. Using our high performance gel permeation chromatography (HP-GPC) (Waters; Mississauga, ON, Canada), the extract of spiked soil samples was cleaned up on two Envirogel columns (19 mm × 150 mm and 19 mm × 300 mm), which were connected in series. DCM was used as mobile phase at a flow rate of 5 mL/min. The target compounds were eluted from 11 to 14 min and co-eluted with lipids and a wax like substance, and thus it was evident that determination of the target compounds in soil and sediment samples was not possible using LC-MS/MS. Considering the target compounds strongly adsorbed to soil, SPE was used for cleanup of the extract. Using a silica gel SPE cartridge, during the DCM wash step the target compounds strongly adsorbed on silica gel while most interference were washed out. After that the target compounds could then be eluted out using a solvent mixture of DCM/ACN (40v/60v). To eliminate any remaining interference in the eluent, a d-SPE approach was used for further sample cleanup. Compared to other sorbents such as C18, zirconia-coated silica particles of Supel™ QuE *Z*-Sep sorbents showed more effective sample cleanup, which was a result of the Lewis acid-base interactions, and also a large capability for adsorption of lipids.

To analyze the trace concentrations of the Scotchgard™ fabric protector components in the sample, high sensitivity UHPLC-MS/MS was used. Different LC columns were tested and compared during method optimization process. C18 UHPLC column was found not to be suitable for this purpose as the mass chromatographic peak of main component (S1) of the pre-2002 Scotchgard™ fabric protector was very broad and fronting. However, using a Kinetex C8 column both S1 and S2 showed very sharp peaks with retention times of 3.60 and 3.45 min, respectively, and thus a large improvement in the lowering of the UHPLC-MS/MS detection limit for S1 and S2 were achieved. Fig. 2 shows the extraction ion chromatograms from a farm soil sample from Cambridge, ON, Canada with and without internal standard spiking (Fig. 2a, b).

In spiked farm soil the method recoveries were 101.4% and 66.1% for S1 and S2, respectively (Table 4). Analyte losses mostly occurred due to inefficient extraction from sample. There was no detection of S1 and S2 in blank samples. The regression coefficient (r^2) of the calibration curves for target compounds were >0.99 in the range from 0.01 to 1000 ng/mL. With the optimized UHPLC-MS/MS method the MLODs and MLOQs of S1 and S2 were at sub-ppb levels (Table 4).

Table 3
The percent recoveries and matrix effects (MEs) of the major components in the Scotchgard™ fabric protectors (S1 and S2) from spiked agricultural soil samples using an ultrasonication extraction method with different solvents and by UHPLC-MS/MS.

Extraction solvent	Pre-2002 (S1) Recovery (%)	Post-2002 (S2) Recovery (%)	Pre-2002 (S1) ME (%)	Post-2002 (S2) ME (%)
DCM	94	31	56	29
Acetone	103	41	50	24
MeOH	107	34	60	28
ACN	125	9	60	33
Formic acid/MeOH (1v/99v)	82	34	64	25
Formic acid/ACN (1v/99v)	106	34	57	23
Hexane/acetone (50v/50v)	101	66	50	25

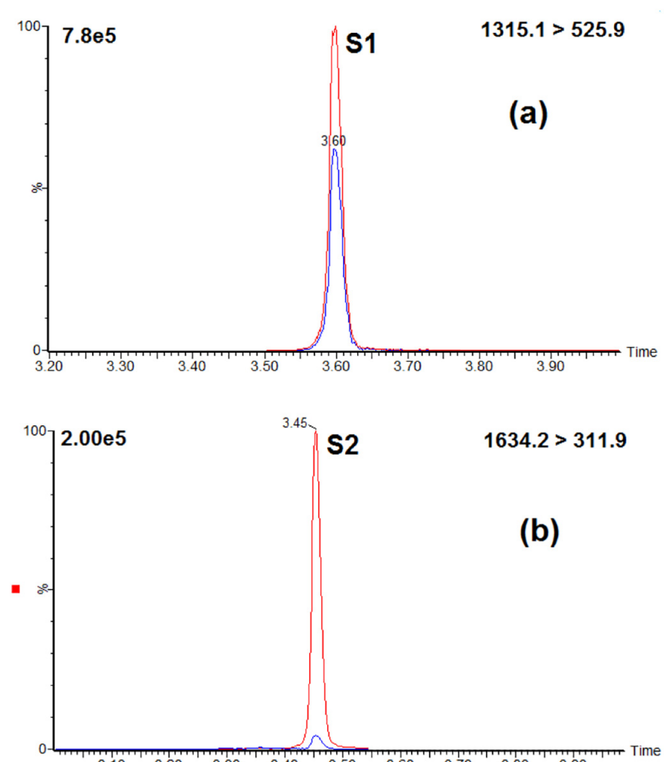


Fig. 2. UHPLC-ESI(-)-QQQ-MS/MS reconstructed mass chromatograms of the main components (a: S1 and b: S2) in the of Scotchgard™ fabric protectors in a representative agricultural soil sample from Cambridge, Ontario, Canada, with (red line) and without (blue line) internal standard spiked. The standard spiked level was 500 ng/mL for each of Scotchgard™ fabric protector standard solution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
The method limits of detection (MLODs), method limits of quantification (MLOQ), percent matrix effect (ME) and recoveries for the determination of the main components of Scotchgard™ fabric protectors by UHPLC-MS/MS.

Product	Major component	MLOD	MLOQ	ME[a]	Recovery
		(ng/g d.w.)	(ng/g d.w.)	(%)	(%)
Scot. pre-2002	S1	0.01	0.03	50.2 ± 4.7	101.4 ± 6.7
Scot. post-2002	S2	0.04	0.09	24.5 ± 4.9	66.1 ± 14.8

[a] Standard addition procedure was be used to measure and calculate ME according the method reported by Rodil, R. et al. Anal. Chem. 2005, 77, 3083–3089.

To confirm the results from the UHPLC-MS/MS analysis, a standard solution and some of the sediment and soil sample fractions were re-analyzed by LC-Q-TOF-MS. For the standard solutions of pre- and post-2002 Scotchgard™ fabric protector products, in their respective mass chromatograms the main detectable component peaks, which contains a polyfluoroalkyl group, were at retention times of 15.8 min (S1) and 14.5 min (S2), respectively, on a Luna C8(2) column (50 mm × 2.0 mm, 3 μm particle size; Phenomenex Com., Torrance, CA, USA). Some other components, which have similar molecular structures with S1 or S2 but with low concentrations, and other non-fluoride surfactants were also detected in the standard solution. The main component of pre-2002 product (S1) had a mass spectral peak at m/z 1315.0591, while that of the post-2002 product (S2) had a mass spectral peak with m/z 1634.3120. The product mass spectrum of m/z 1315.0591 (S1) had two main ions with m/z 418.9713 and 525.9792 (Fig. 3a), which correspond to fragment ions of $[C_8F_{17}]^-$ ($m/z = 418.9734$) and $[C_8F_{17}SO_2N(C_2H_5)]^-$ ($m/z = 525.9775$), respectively. The product mass spectrum of m/z 1634.3120 (S2) had two main ions with m/z 218.9860 and 311.9751 (Fig. 3b), which corresponded to the fragment ions of $[C_4F_9]^-$ ($m/z = 219.9862$) and $[C_4F_9SO_2N(CH_3)]^-$ ($m/z = 311.9746$), respectively. This information is consistent with the voluntarily phase-out by the 3 M Company of PFOS related products by the end of 2000 (The 3M Company, 2000). It was not possible to determine the exact molecular structures of S1 and S2, i.e. the polymer backbone, with the present mass spectral information. Also, the polymer composition is withheld as a trade secret by the 3M Company. Combining with the high mass resolution ToF-MS date and information in the literature, we confirmed that S1 and S2 are both C8 and C4, respectively, side-chain fluorinated polymers. Also, we know that the non-fluorinated polymer backbones in S1 and S2 are *co*-polymers with a short chain because no repeat units composed of monomer mass was observed from the MS or MS/MS analysis. The molecular structures of S1 and S2 are very different (i.e. a C_8F_{17} group in S1 as opposed to C_4F_9 group in S2), and also their non-fluorinated polymer backbones are different. Finally, no mass spectral information also supports that S1 and S2 are N-alkyl perfluoroalkyl sulfonamidoethyl acrylate- and methacrylate-based polymers, although this type of covalent bond in the molecular structure is commonly used for side-chain fluorinated polymers (Buck et al., 2011; Liu and Avendano, 2013). Neither the $[C_4F_9SO_2N(CH_3)CO_2]^-$ nor the $[C_8F_{17}SO_2N(C_2H_5)CO_2]^-$ ions were detected in their standard solutions.

The components of S1 and S2 were also detected in the sediment and soil sample fractions after re-analyzing by LC-Q-TOF-MS. Their retention times and mass spectra (MS and MS/MS) matched well with those of the standards, and thus confirmed the results of the UHPLC-MS/MS analysis. Fig. 3(c, d) shows the product mass spectra of precursor ion with m/z 1315.0591 and m/z 1634.3120 at retention time of 15.8 and 14.5 min in mass chromatogram of a representative sediment sample from western Lake Erie (Table 1).

LC-ESI-MS/MS analysis can be affected by sample-based matrix effects (Ferguson et al., 2000). We used a standard addition method to measure matrix effects (MEs) (Rodil et al., 2005). Even for samples that were cleaned up using SPE and d-SPE, there was significant signal suppression, especially for S2 and the percent matrix effect (ME) value was ≪100% (Table 4). As there is no suitable internal standard available for S1 and S2 quantification, a standard addition method was used. In this method any signal suppression from matrix was corrected by the spiked standards because the target compounds suffered the same matrix effect in the MeOH spiked and standard solution spiked samples. A disadvantage is that each sample had to be analyzed at least twice by UHPLC-MS/MS and the spiked concentration of standard (C_s) had to be at a comparable level as in the sample (C_x).

3.2. Main components of pre- and post-2002 Scotchgard™ fabric protectors in sediment and soil samples from the Great Lakes basin

The concentrations of the main components (S1 and S2) in the Scotchgard™ fabric protector products in the present sediment samples from western Lake Erie and Lake Huron are listed in Table 5. In all of the samples S1 was quantifiable and ranged from 0.24 to 461.59 ng/g d.w. In 80% of the samples S2 is quantifiable and ranged from <0.04 to 24.08 ng/g d.w. The greatest concentrations of S1 and S2 were found in samples from western Lake Erie (No. 13 and No. 14), followed by two samples from Saginaw Bay (No. 1 and No. 3) with S1 concentrations of 167.14 and 129.00 ng/g d.w. The concentrations of S1 and S2 in sediment samples that were essentially sand substrate (Nos. 2, 5, 6 and 7) were relatively lower (ND to 4.63 ng/g d.w.) as compared to the sediment samples with a silt consistency (Table 5). The greatest concentration of S1 and S2 in the western Lake Erie sediment samples is likely due to their close proximity to the Detroit River in-flow. It is well established that the Detroit River outflow is the major contamination source/vector of organic pollutants into western Lake Erie, where relative to the central and eastern basins of Lake Erie, much higher PCB, PBDE and even bisphenol A (BPA) concentrations have also been reported in surficial sediment samples (Marvin et al., 2013; Lu et al., 2015). The high concentrations of S1 and S2 detected in sample site No.1 from Saginaw Bay may be the result of the Au Gres River outflow which drains into Lake Huron.

In all of the agricultural soil samples from the southern Ontario field sites, S1 was detectable while in 62% of the soil samples S2 was detectable. Among all the soil samples, the greatest S1 (mean 236.36 ng/g d.w.; range from 41.87 to 622.46 ng/g d.w) and S2 (mean 4.12 ng/g d.w.; range from 0.84 to 10.39 ng/g d.w.) concentrations were in the soil samples from Cambridge, Ontario, which were the sites augmented with biosolid in the previous fall 2013 (Table 5). Since the Cambridge sites were the only sites augmented with biosolids, this indicates that the elevated S1 and S2 concentrations are sources from the wastewater treatment plants from which the biosolids came. Other factors have to be considered as well such as their S1 and S2 adsorption behaviour and degradation might have influenced the distribution patterns in soil from the studies sites.

3.3. Fluorinated copolymer surfactants and other PFASs concentrations in the sediment and soil samples

In addition to S1 and S2, all of the sediment and soil samples were also analyzed for thirteen PFCAs, five PFSAs and four perfluoroalkyl sulfonamides using our previously reported method (Chu et al., 2016). In all samples, FBSA, FOSA, N-MeFOSA and N-EtFOSA were not detectable, which are considered as possible intermediates in the degradation of S1 and S2. The dominant compounds in the soil and sediment samples are PFOS, perfluoro-*n*-octanoic acid (PFOA) and perfluorononanoic acid (PFNA) (Table 5).

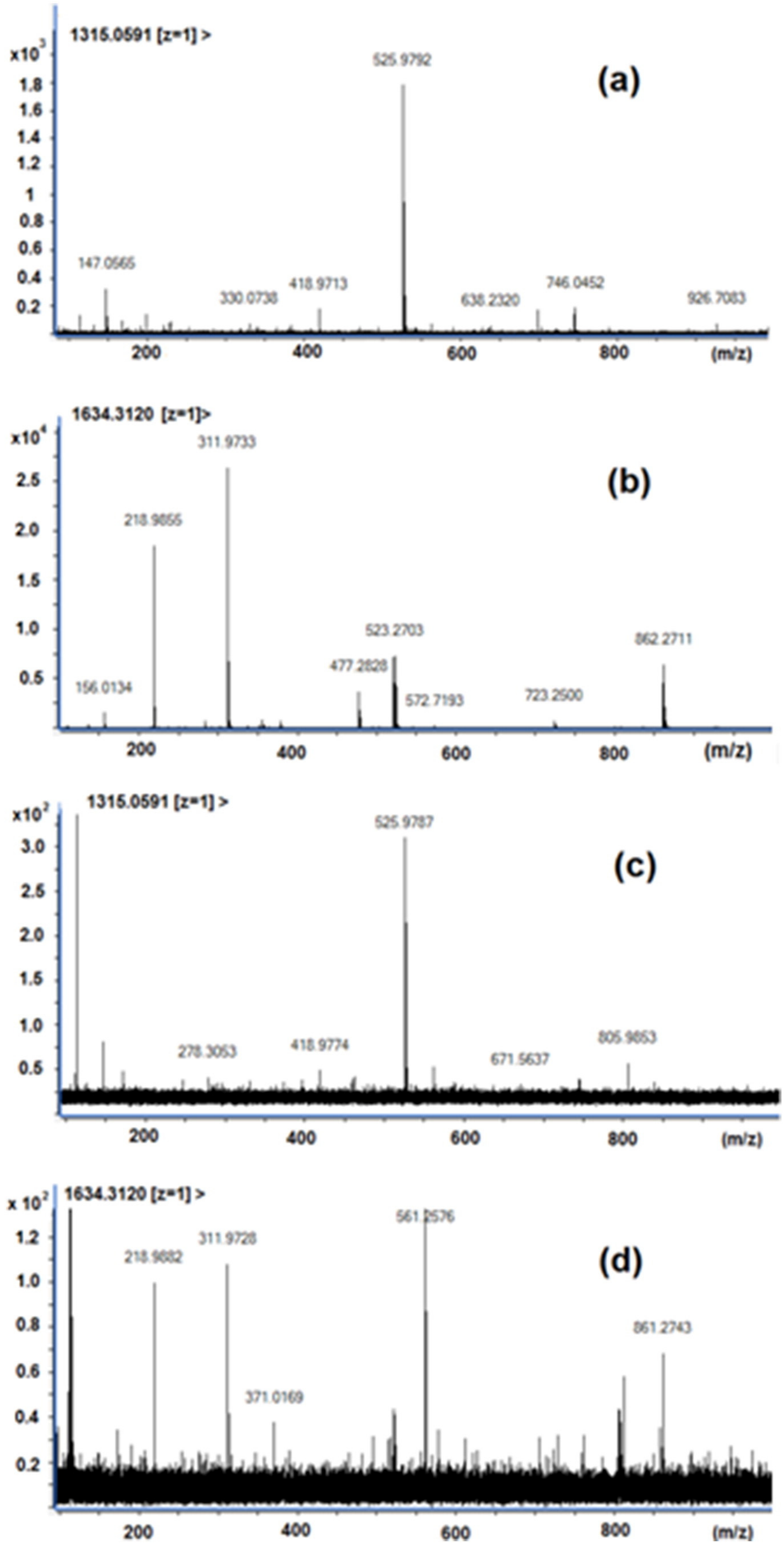


Fig. 3. Product mass spectra of the precursor ion with *m/z* 1315.0591 and 1634.3120 (a) in the Scotchgard™ fabric protector pre-2002 standard solution, (b) in the Scotchgard™ fabric protector post-2002 standard solution, and (c, d) in a sediment sample from Lake Erie (site 14), Canada, analyzed by LC-ESI(-)-Q-ToF (MS/MS).

Table 5
Concentrations of the main components in the Scotchgard™ fabric protectors (S1 and S2) and other major PFASs present in the sediment and agricultural soil samples (unit: ng/g d.w.).

Sample No.	Sample location[a]	S1	S2	PFOA[b]	PFNA[b]	PFOS[b]	$\sum_{22}$PFASs
Sediment							
1	Saginaw Bay, Lake Huron	167.14	2.14	0.35	0.21	1.54	3.49
2	Saginaw Bay, Lake Huron	2.08	0.23	ND	ND	ND	<MLOQ
3	Saginaw Bay, Lake Huron	129.00	2.20	<MLOQ	0.26	1.58	2.53
4	Saginaw Bay, Lake Huron	17.79	0.10	ND	ND	0.22	0.23
5	Saginaw Bay, Lake Huron	0.24	ND	ND	ND	1.25	1.41
6	Saginaw Bay, Lake Huron	0.59	ND	ND	ND	<MLOQ	<MLOQ
7	Saginaw Bay, Lake Huron	4.63	0.78	ND	ND	ND	<MLOQ
8	Lake Huron	33.12	0.11	0.75	0.61	3.47	5.71
9	Lake Huron	9.46	ND	0.18	0.18	0.63	1.87
10	Lake Huron	29.89	0.10	0.33	0.94	2.90	6.91
11	Lake Huron	27.76	0.13	0.44	0.32	1.38	4.78
12	Lake Huron	2.11	<MLOQ	<MLOQ	0.07	0.27	1.04
13	Lake Erie	461.59	14.66	ND	ND	0.20	0.29
14	Lake Erie	225.19	24.08	ND	ND	0.29	0.38
15	Lake Erie	0.62	0.25	ND	ND	ND	<MLOQ
Farm soil		Mean					
TI (n = 9)	Tillsonburg, ON, Canada	3.02	0.16	ND	<MLOQ	<MLOQ	<MLOQ
DE (n = 1)	Delhi, ON, Canada	17.36	0.52	<MLOQ	ND	ND	ND
CB (n = 3)	Cambridge, ON, Canada	236.36	4.12	ND	<MLOQ	2.01	2.42

[a] See Table 1 and Fig. 1 for details regarding the samples and sampling locations.
[b] See SI table for the full chemical name for these abbreviations in refer. Chu and Letcher, 2014.

In the sediment samples the $\sum_{22}$PFAS concentrations (mean 1.94 ng/g d.w.; ranged from <MLOQ to 6.91 ng/g d.w.) were comparable to S2, but orders of magnitude lower than S1 (Table 5). These PFOS, PFNA and PFOA results are very comparable with other reports of these PFASs in sediments (White et al., 2015; Yeung et al., 2013; Becanova et al., 2016; Long et al., 2013; Thompson et al., 2011; Beskoski et al., 2013). In the sample group from Lake Huron the S1 to $\sum_{22}$PFAS concentration ratio was 5.7; however, for some samples with high concentrations of S1 the ratio was much greater. In the two samples from western Lake Erie (Nos. 13 and 14) the ratios were 1616 and 589, respectively. Codling and et al. (2014) measured total fluorine (TF), extractable organic fluorine (EOF) and poly- and per-fluorinated compounds abbreviated as PFCs (rather than the correct PFASs), and with exception of polymers, in eight dated cores of sediment taken along with 27 surface sediments from Lake Michigan in 2010. The results showed that when accounting for any quantified PFCs, they represented only 0.16% and 0.09% of EOF in sediment cores and surface sediment samples, respectively. In another environmental investigation for PFASs in sediment, it was reported that the known PFCAs and PFSAs accounted for 2 to 44 % of the EOF (Yeung et al., 2013). It was suggested that the unidentified organic fluorine shared similar sources of contamination or physicochemical characteristics to those of PFCAs and PFSAs, and it was hypothesized that sediments might contain some precursor and/or intermediate compounds of commonly measured PFSAs (Codling et al., 2014; Yeung et al., 2013). Our present results provide evidence to support this phenomenon. For example, in sample No.13, the ratio of fluorine in $\sum_{22}$ PFASs versus fluorine in S1 was only about 0.8%. Aside from the reported PFASs, a larger number of other fluoropolymers are likely to be present in the environment (Liu and Avendano, 2013).

The target PFASs and S1 and S2 results for the soil samples are listed in Table 5. The $\sum_{22}$PFAS concentration in the sample from Cambridge, ON, Canada was significantly higher than in those from Tillsonburg and Delhi. The concentration was also was comparatively greater than $\sum_{22}$PFASs in the soil samples collected from the location where there was no evident human impact (Zareitalabad et al., 2013; Rankin et al., 2016). This is consistent with that high concentration of S1 and S2 found in these same samples from Cambridge. This result suggests that these PFASs are mainly from sewage biosolids, which are used for soil amendment in the area in recent years, which is similar to related studies that have been published recently (Washington et al., 2010; Strynar et al., 2012; Arvaniti and Stasinakis, 2015; Kim et al., 2016).

3.4. Possible sources of Scotchgard components and sediment and soil

Scotchgard™ fabric protector components found in the present sediment and soil samples demonstrated the presence of side-chain fluoropolymers in terrestrial and aquatic environments of the Great Lakes, which to our knowledge is the first such report. WWTP sludge is considered as a main PFAS pollution source in aquatic systems because high concentrations of PFASs have been reported (Yeung et al., 2013; Arvaniti and Stasinakis, 2015; Kim et al., 2016; Alder and van der Voet, 2015; Lee et al., 2014). A potential source of these fluoropolymers in the environment is also via WWTP discharge into water systems or via redistribution of biosolid slurry for crop field applications. Different from other hydrophobic contaminants such as polychlorinated biphenyls (PCBs), these fluoropolymers exhibit both hydrophobic and lipophobic characteristics. A simple hydrophobic partitioning paradigm was found not to be suitable when applied to calculate the sorption of these compounds to sediment and sludge (Buck et al., 2011; Zareitalabad et al., 2013; Ololade et al., 2016; Rankin, 2015). High concentration of S1 component and low concentration of its potent degradation products (FOSA, N-EtFOSA and PFOS) found in the present sediment and soil samples implies that these side-chained polymers are likely to be resistant to biodegradation. This present result helps to explain why known PFCAs and PFSAs account for low percentages of the extractable organic fluorine (EOF) in sediment. More environmental investigation is necessary for these components and other related fluoropolymers to better understand their distribution, transforming mechanisms, degradation and potential impact on the environment and human health.

Acknowledgements

This study was funded by the Ecotoxicology and Wildlife Health Division (Environment and Climate Change Canada (ECCC)) and ECCC's Chemicals Management Plan (CMP). We thank ECCC colleagues Daryl McGoldrick and Sean Backus for sampling of the surficial sediments from the western Lake Erie sites, and Pamela Martin and Glenn Barrett for collection of the southern Ontario soil samples. We thank Elizabeth Murphy, Elizabeth Hinchey Malloy and Glenn Warren at the US EPA Great Lakes National Program Office, and all individuals on The Guardian, as well as Michelle van den Brand at the Great Lakes Environmental

Center (GLEC; Michigan Department of Natural Resources) for sampling of the surficial sediments from the Lake Huron sites.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.scitotenv.2017.06.252.

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